

TEKAMAH, NE, USA

WMO: 725527

Lat: 41.764N

Lon: 96.178W

Elev: 1028

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94978

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.0	-0.5	-13.0	2.8	-4.7	-8.5	3.6	-0.4	34.8	21.4	31.6	22.8	7.4	320	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	94.0	75.3	91.2	74.7	88.8	73.6	79.8	88.9	77.9	87.5	76.2	85.3	14.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	147.0	85.7	75.0	136.3	83.9	73.1	127.7	81.8	44.1	88.7	42.2	87.7	40.3	85.2	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.6	25.5	22.7	DB	-14.9	99.4	3.9	3.0	-17.7	101.5	-20.0	103.3	-22.3	105.0	-25.1	107.2	(4)
(5)			WB	-15.3	83.1	3.9	1.4	-18.1	84.2	-20.4	85.0	-22.6	85.8	-25.4	86.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	22.2	26.0	38.7	51.0	62.4	72.4	76.0	72.7	65.6	52.2	38.6	25.9	(6)	
	DBStd	21.22	12.21	12.52	11.90	9.96	8.82	6.64	5.74	5.40	8.07	9.65	10.22	11.47	(7)	
	HDD50	3234	863	673	385	104	8	0	0	0	1	88	363	749	(8)	
	HDD65	6353	1328	1092	815	431	156	15	2	6	91	410	793	1214	(9)	
	CDD50	3393	0	1	36	134	394	671	807	704	469	157	20	0	(10)	
	CDD65	1038	0	0	2	11	77	236	345	245	108	14	0	0	(11)	
	CDH74	10972	0	0	35	215	904	2491	3661	2249	1205	205	7	0	(12)	
	CDH80	4364	0	0	7	68	337	1021	1613	826	434	57	1	0	(13)	
(14) Wind	WSAvg	10.0	10.8	11.0	11.8	12.7	11.8	10.0	7.4	6.6	7.9	9.7	10.4	10.4	(14)	
(15) Precipitation	PrecAvg	28.4	0.7	0.7	2.0	3.0	4.1	4.1	3.1	3.3	3.1	2.0	1.3	0.7	(15)	
	PrecMax	41.9	2.3	2.8	6.8	8.4	11.3	10.9	8.5	9.2	7.7	7.0	4.4	2.7	(16)	
	PrecMin	14.9	0.0	0.0	0.0	0.4	1.3	0.7	0.8	0.4	0.2	0.0	0.0	0.0	(17)	
	PrecStd	6.5	0.5	0.6	1.5	1.9	1.9	2.1	1.6	2.1	1.9	1.6	1.1	0.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.7	63.3	78.8	87.9	93.3	97.3	98.7	95.7	93.0	86.7	73.2	57.3	(19)	
		MCWB	44.7	51.4	58.4	63.5	69.3	74.6	77.7	76.4	72.0	67.8	57.4	48.0	(20)	
	2%	DB	48.2	54.1	71.3	80.6	88.1	92.7	94.6	90.9	89.5	79.6	66.1	51.0	(21)	
		MCWB	40.5	46.5	56.8	61.2	68.1	73.6	77.3	76.9	71.6	63.3	53.8	44.0	(22)	
	5%	DB	42.6	47.7	63.6	74.1	83.5	89.7	91.3	88.0	84.9	74.5	60.6	45.6	(23)	
		MCWB	37.0	42.1	53.6	57.5	65.2	72.0	77.2	75.7	70.1	60.5	50.3	40.1	(24)	
	10%	DB	37.6	42.4	57.1	69.3	79.1	86.2	88.4	84.4	81.3	70.0	54.9	40.8	(25)	
		MCWB	34.2	37.8	48.3	55.4	63.3	70.8	75.7	73.4	67.6	57.7	46.8	36.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	53.8	63.4	66.1	74.3	79.5	83.2	81.2	77.0	70.7	60.3	51.5	(27)	
		MCDB	55.9	59.6	73.4	81.0	86.4	90.4	91.5	89.1	87.5	82.0	69.1	54.8	(28)	
	2%	WB	41.2	46.2	59.1	62.9	70.9	77.1	80.7	78.7	74.5	66.5	55.8	44.2	(29)	
		MCDB	48.0	52.5	66.5	76.0	82.7	87.7	89.3	87.7	84.2	75.0	64.5	49.5	(30)	
	5%	WB	37.5	41.8	54.2	59.8	68.2	75.1	78.7	77.0	72.5	63.0	51.4	40.5	(31)	
		MCDB	41.6	47.6	63.6	71.9	79.2	85.4	88.0	85.6	81.5	71.1	58.9	45.2	(32)	
	10%	WB	34.6	38.1	49.1	56.7	65.5	73.2	77.0	75.0	70.4	59.2	47.2	37.1	(33)	
		MCDB	37.6	41.8	57.6	67.0	75.6	83.6	86.5	82.6	78.6	68.2	54.7	40.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	18.8	21.7	23.9	22.5	21.8	21.4	21.5	25.2	24.6	22.0	18.1	(35)	
		MCDBR	26.5	26.1	33.0	33.7	29.3	25.6	24.0	24.9	28.9	31.8	31.7	25.3	(36)	
	5% WB	MCWBR	19.6	18.6	19.7	17.8	12.9	11.1	11.4	12.6	13.4	16.7	20.0	18.8	(37)	
		MCWBR	24.1	24.6	29.5	28.1	24.3	21.5	21.3	21.7	23.8	25.8	28.8	23.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.288	0.314	0.335	0.395	0.419	0.422	0.432	0.422	0.380	0.334	0.299	0.288	(40)	
		taud	2.417	2.366	2.378	2.227	2.229	2.284	2.253	2.281	2.356	2.472	2.494	2.438	(41)	
	Ebn at Noon		274	283	289	277	272	271	267	266	270	272	269	264	(42)	
		Edh at Noon	25	30	34	43	44	42	43	40	35	27	23	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		624	936	1205	1547	1829	2036	2071	1767	1448	1008	702	521	(44)	
	RadStd		59	60	105	115	135	157	125	124	93	120	46	49	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page